

Facilitator copy. This manual contains answer guides, rubrics, grading thresholds, and standards codes. **Do not publish it to students.** It lives in the teacher's files only — not inside the Canvas course scholars can see.



Optima Academy Online
An Education Experience Company

TEACHER MANUAL · FACILITATOR GUIDE

Chemistry I Honors

Florida course 2003350 · Grades 10–12 · Full year · Fully asynchronous (also taught live) · Getting Started + 9 modules · 76 lessons · ~500:1 student–teacher ratio

This manual documents Chemistry I Honors *as it was actually built*. Every module, lesson, assessment, rubric, and standard below is read from the finished course — the course-identity file, the scope-and-sequence plan, the built module folders, and the four quarterly standards crosswalks. It exists to hand you everything you need to teach roughly 500 asynchronous scholars without re-reading the whole course. Where a detail depends on your Canvas instance or your section, it is flagged [VERIFY].

- | | | | |
|---|-------------------------------------|----|-------------------------------------|
| 1 | How this course works | 8 | Coach, not catch |
| 2 | Ungraded interactives & wrappers | 9 | Weekly announcement scripts |
| 3 | AI & VR (student-facing) | 10 | Teacher workflow & workload map |
| 4 | Using Ignite AI to support scholars | 11 | Standards coverage map |
| 5 | Grading categories & gradebook | 12 | What makes this course classical |
| 6 | Rubrics for teacher-graded items | 13 | Sensitive-content flags & scripts |
| 7 | Assessment & activity types | 14 | Adapting to a live 2-day/week class |

1. How This Course Works

Three quick walk-throughs so you know exactly what you are facilitating.

A typical lesson

Each lesson is one self-contained Canvas page a scholar reads on their own. Top to bottom, a lesson page carries:

1. A navy header + metadata strip — quarter/module, lesson number, topic, and the quarter's Grand Question.

2. **Learning objectives** — 2–4 "I will be able to..." statements.
3. **Reading prose in Optima voice** — the core content, written into the page and organized as an empirical-questioning arc (observe → compare → find patterns → explain → evaluate evidence → extend to current research).
4. **A history-of-science callout** (blue) and/or **current-research callout** (gold), where they fit the topic.
5. **An embedded interactive** on some lessons — a PhET/Concord simulation or a custom widget — to manipulate the concept. Not every lesson has one.
6. **"Your VR This Week"** — a green pointer connecting the lesson to the week's VR activity and the module's cumulative VR project.
7. **A short Reading Check** — a separate 3–5 question auto-graded Canvas quiz. This is the only graded touchpoint on the lesson itself.

A typical week

A normal content week runs **2–3 lessons** (3 for descriptive stretches, 2 for calculation-heavy ones), plus one quantitative assignment and one VR activity. Nothing here costs more than a few minutes of your time — it is all auto- or self-graded.

WHAT THE SCHOLAR DOES	GRADED?	TIER	YOUR ROLE
Reads 2–3 lesson pages	—	—	None; content is self-contained
Takes a Reading Check per lesson	Yes	Tier 1	Auto-graded; glance at low scores
Completes one hands-on quantitative assignment	Yes	Tier 1	Auto / self-check; Excel where flagged
Completes one VR activity (or non-VR alternate)	Yes	Tier 4	Completion; spot-check submissions
Advances the module's cumulative VR project	At module end	Tier 3	Grade the final video once per module

Major assessments

Per module, exactly **two items reach your desk**: the writing assignment and the VR project video. Everything else is auto-graded. Two cumulative exams anchor the year.

ASSESSMENT	WHERE	TIER	GRADING TIME	WHAT IT ASKS
Reading Checks (~76)	Every lesson	1	~0	3–5 MC/TF/matching comprehension items
Weekly hands-on (~33)	Every week	1	~0	Numeric/MC problem sets; data analysis
Module checkpoints (1–2 / module)	Mid-module	auto	~0	Pace check drawn from reading-check style
Module exams (9)	Module end	auto	~0	Full-module, auto-gradable item bank
Module writing (9)	Module end	3	~5 min/scholar	CER, lab report, or scientific argument
Module VR project video (9)	Module end	3	~5 min/scholar	Recorded explanation of the module's chemistry
Semester 1 exam	After M4	auto	~0	Cumulative M1–M4
Final exam	After M9	auto	~0	Cumulative; Q3–Q4 emphasis

2. Ungraded Interactives & the Module Wrappers

Ungraded interactive activities cost you nothing

Many lessons embed a simulation or a self-check widget — a PhET "Build an Atom," a custom average-atomic-mass calculator, a Lewis-structure checker, a dimensional-analysis trainer, an ICE-table solver. These live on GitHub-hosted pages inside the lesson iframe, give the scholar **immediate feedback**, and produce **no submission**. They are formation-by-practice, not busywork to chase. A scholar who skips them does not show up as a missing grade — they show up indirectly, in weaker reading-check and hands-on performance. Do not try to track interactive completion; watch the auto-graded results instead.

MODULE INTRO / OUTRO WRAPPERS ARE SURVEY-MODE QUIZZES, NOT PAGES

Each module opens and closes with a **Canvas Quiz set to survey mode** (not a wiki page). They capture the scholar's first and refined answer to the module's Grand Question, a confidence rating, "what's still unclear," and a short virtue reflection — as **sortable cohort data**.

The single highest-leverage move in the course: spend 15–20 minutes per module reading the *outro* responses. Sort for low-confidence clusters and repeated "still unclear" concepts, and you will spot confusion and at-risk scholars before the exam does. Act on it with one targeted announcement, an Ignite AI nudge, or an individual message.

3. AI & VR in This Course (Student-Facing)

AI — instrument, not oracle

The course's posture, written into its lessons as the **AI-as-Instrument thread**, is *AI awareness, not required tool use*. Once per quarter a lesson pairs a historical instrument with its modern AI parallel and insists on **human verification against primary evidence**:

- **Q1** — Microscope/STM → AI imaging of atomic-scale structure; verify against measured data.
- **Q2** — Spectroscopy → AI pattern recognition of spectra; verify against known spectra.
- **Q3** — Instruments & datasets → AI data analysis of gas-law/calorimetry data; verify against the scholar's own calculations.
- **Q4** — Scientific literature → AI research tools; verify claims against primary evidence.

No lesson requires a specific AI tool, and no callout persuades. You can speak to AI use anywhere it comes up; do not add new AI requirements.

VR — build, then explain

Each content week has one 30–60 minute VR activity on Optima's VR campus (a build, a guided exploration, or an apply-your-knowledge station). Across the module, scholars advance one **cumulative VR project** and, at the end, **record themselves inside VR explaining the chemistry** the build shows — the assessed work is the spoken explanation, not a static model. **Every VR activity ships with a non-VR alternate** (screen-recorded or narrated) that meets the same objectives; accept it without penalty. You review the final videos and can assemble a gallery walk of captures for the class.

[VERIFY] Optima is moving between VR platforms. This manual describes VR by what the scholar does; confirm the current platform's capture-and-submit steps against your instance and share those with scholars.

4. Using Ignite AI to Support Scholars (Teacher-Facing)

[VERIFY] before relying on any feature. Ignite AI's exact feature names and availability vary by Canvas release and by what Optima has enabled. Confirm the current feature set against Optima's Canvas instance; the capabilities below are described functionally.

At 500:1, the Canvas AI assistant is how you make grading and outreach feel personal without adding hours. Use it, functionally, to:

- **Draft individualized feedback** on a writing assignment or VR video that you then edit and own — faster Tier 3 turnaround, same human judgment.
- **Summarize the outro-wrapper analytics** for a module to find the cohort's sticking points fast (see the triage move in Section 2).
- **Surface scholars who need a nudge** — low outro confidence, missed work, a declining reading-check trend — and draft a warm check-in message.
- **Generate practice variants** for a scholar who needs another rep on a Tier 1 skill (a second set of mole conversions, a fresh Lewis-structure, another half-life problem).

- **Communicate with families** in clear, kind summaries, and translate messages when that helps.

Standing rule: AI drafts, the teacher decides. You read, edit, and own anything sent to a scholar or family. AI is never used to surveil, rank, or auto-penalize (see Section 8).

5. Grading Categories & Canvas Gradebook Setup

Four assignment groups map to the tier system and to the weights printed on the student syllabus. ([**VERIFY**]) confirm these weights against your section's official gradebook before term start; adjust here if your section differs.

ASSIGNMENT GROUP (CATEGORY)	WEIGHT	TIER	GRADING METHOD	CONTENTS
Weekly Practice & Comprehension	35%	1	Auto	Reading Checks + weekly hands-on quantitative sets
VR Activities & Projects	20%	4 / 3	Completion + rubric	Weekly VR (completion) + module VR project videos (rubric)
Checkpoints & Exams	30%	auto	Auto	Module checkpoints + module exams + 2 cumulative exams
Module Writing	15%	3	Rubric	CER / lab report / scientific argument (1 per module)

STEP-BY-STEP CANVAS SETUP

1. **Create the four assignment groups** above (Gradebook > Assignments > + Group).
2. **Turn on group weighting** and enter 35 / 20 / 30 / 15.
3. **Set the grading scheme** to the school's standard letter scale.
4. Mark the intro/outro wrappers as survey / "Do not count toward final grade" (or a small completion score) so reflection data does not distort grades.
5. **Set late/missing policy** to match the On-Demand rules: no per-due-date penalty, zero for work not done by quarter's end, 10% deduction in a quarter's final 14 days. ([**VERIFY**]) live sections may weight participation instead.
6. **Use the Module Requirements / progress view** as your at-a-glance completion signal across the ungraded scaffolding.

6. Rubrics for Every Teacher-Graded Item

Two kinds of item reach you each module. Paste each rubric into the Canvas item as a **native Canvas Rubric** so it scores in SpeedGrader. All rubrics keep to four criteria and stay identical across modules, so grading holds at ~5 min/scholar.

6a. Standard writing rubric (all 9 module writing assignments)

CRITERION	FULL CREDIT	PARTIAL	MINIMAL
Understanding of the chemistry	Concepts are correct and complete; terminology precise; data/structures right	Mostly correct; a minor conceptual slip or gap	Key concept missing or wrong
Quality of reasoning	Claim follows from evidence; each step justified; counter-case addressed where asked	Reasoning present but a link is thin or unsupported	Assertion without evidence, or a logic break
Clarity & precision of expression	Organized, readable; correct units, sig figs, symbols, and equations	Understandable; some sloppiness in units/format	Hard to follow; pervasive unit/format errors
Intellectual honesty	Reports real data/limits; cites sources; own voice; AI use disclosed	Minor gaps in citing or acknowledging limits	Unsupported certainty, or undisclosed borrowed work

Suggested scale: 4 / 3 / 2 / (0–1) per criterion; sum to the item's gradebook value. Lead feedback with what worked, then the single next step (Section 8).

6b. VR project video rubric (all 9 modules, shared)

CRITERION	FULL CREDIT	PARTIAL	MINIMAL
Accuracy & depth of spoken explanation	Explains the module's chemistry correctly and in real depth	Mostly correct; shallow in places	Vague or contains errors
Recognizable accuracy of the 3D build	Model is correct at a high-school level and clearly referenced on camera	Model roughly right; loosely tied to the talk	Model wrong or barely used
Evidence of week-by-week iteration	Build clearly grew across the module's weeks	Some growth visible	Assembled at the last minute
Clarity of communication	Well-paced, organized, audible narration	Understandable but disorganized	Hard to follow

Identical criteria apply to the non-VR alternate. Grade the *explanation*, not model polish.

6c. Per-module writing prompt & full-credit shape

MOD	TYPE	PROMPT & WHAT A FULL-CREDIT ANSWER MUST CONTAIN
1	Lab report	Density of an unknown material: reports the measurement, distinguishes precision from accuracy, and computes percent error, with a data table and a conclusion that names error sources.
2	Scientific argument	"How the Atom Changed": uses cathode-ray, gold-foil, <i>and</i> atomic-spectra evidence to argue the Thomson→Rutherford→Bohr revision; must include a counterclaim and reasoning on why models change.
3	CER	Predicts a molecule's polarity and justifies it from bonding type <i>and</i> molecular geometry (VSEPR); correct terminology; addresses a symmetric-molecule counter-case.
4	Lab report	From a limiting-reactant investigation: correct stoichiometric calculations, theoretical and percent yield, and an account of the gap between them.
5	Lab report / CER	From calorimetry data: a correct $q = mc\Delta T$ calculation, an enthalpy interpretation, and reasoning about energy conservation in the system.
6	Scientific argument	Explains a macroscopic property (boiling-point order, or why road salt melts ice) from intermolecular forces / colligative behavior, tied to a real-world case.
7	CER	Applies Le Châtelier's principle to a real system (ocean acidification or the Haber process): predicts and justifies the response to a stress, with a societal connection.
8	Scientific argument	Argues which battery/cell chemistry fits a stated use case, grounded in cell potential and redox, and weighs costs and benefits.
9	Case study + CER	Analyzes Curie's radioactivity work, then makes an evidence-based argument about a modern nuclear application (energy or medical isotopes), weighing benefits and risks with a responsible-use frame.

Free-response quiz items. By design, Reading Checks, checkpoints, and module exams are built to be *auto-gradable* (MC / TF / matching), so the only items you score by hand are the two above. If you convert any exam item to free response, grade it with a compact 2–3 point guide: full credit = the key idea from that lesson stated correctly; partial = right idea, incomplete; none = missing or wrong. Keep such guides in this teacher copy only.

7. Assessment & Activity Types — How to Run & Grade

TYPE	SCORED	WHAT IT IS & YOUR MOVE
Reading Check	Auto	3–5 MC/TF/matching per lesson; every Q1–Q2 check now carries at least one higher-order item beyond recall. Move: none; scan the item analysis if a whole cohort misses one.
Weekly hands-on	Auto / self-check	Numeric or MC problem set; Excel data analysis where flagged. Move: none; note recurring wrong answers for your announcement.
Weekly VR activity	Completion	One-sitting build/exploration, with a non-VR alternate. Move: mark complete; spot-check a few captures.
Module checkpoint	Auto	Mid-module pace check (two in the large modules M2, M4, M7, M8). Move: watch who stalls here.
Module exam	Auto	Full-module item bank with an item-alignment map. Move: none beyond review.
Module writing	Rubric (you)	The Tier 3 piece. Move: batch-grade with rubric 6a; ~5 min each.
VR project video	Rubric (you)	The Tier 3 piece. Move: batch-grade with rubric 6b; consider a gallery walk of the best.
Cumulative exam	Auto	Semester 1 (M1–M4) and Final (Q3–Q4 emphasis). Move: post a review announcement first.

No graded discussion boards. This async, 500:1 course carries its argumentation load in the writing assignments and the VR videos, not in threaded discussions. If your *live* section adds discussion, grade it on a simple substantive / thin / missing scale and drop 2–3 public model comments rather than replying to every post.

8. Coach, Not Catch

Your job is to move each scholar forward, not to detect and punish. In practice:

- Feedback names the **next step**, not just the error — and leads with what is working.
- Allow re-attempts on Tier 1 practice; reading checks and hands-on sets are learning tools, not traps.
- Use outro analytics and Ignite AI to **find scholars who need help early**, then reach out — not to build a case.
- On suspected AI misuse or copied work, open a conversation: ask the scholar to talk through their thinking or re-derive a calculation. Treat it as formation, not a verdict.
- When you highlight strong work publicly, model good thinking; do not single out weak work for criticism.

9. Weekly Announcement Scripts

Because scholars are asynchronous, the weekly announcement is your main presence. Keep it short and human; fill the bracketed slots.

REUSABLE TEMPLATE

[Module N — This Week] Scholars, you should be in **[lesson(s)]** this week. Focus: *[one key idea]*. Don't miss: **[the due item — hands-on set / VR increment / writing or video]**. One tip: *[the common trip-up]*. Stuck? Come to office hours or message me — that is what I am here for. Keep going. — [name]

- 0 Welcome to Chemistry I Honors! This week is Getting Started: set up OptimaXR, start your science notebook, and practice one Claim–Evidence–Reasoning response. Don't miss the "You're Ready" check. Say hello in office hours — I'd love to meet you.
- 1 We begin with the big question: *what is matter?* Focus this stretch is measurement — units, significant figures, and dimensional analysis. Don't miss the density lab report. Tip: sig figs sink more scholars than the chemistry does; slow down and count.
- 2 The atom's story is our longest argument from evidence. Watch how cathode rays, gold foil, and spectra force the model to change. Don't miss the "How the Atom Changed" argument — it *requires a counterclaim*. Advance your Atomic Model Evolution build each week.
- 3 Now: why matter changes. We move from bonds to molecular shape and polarity. Don't miss the polarity CER. Tip: geometry, not just bond type, decides polarity — draw it before you decide.
- 4 The mole is the spine of everything after this. Balance carefully, then let stoichiometry do the work. Don't miss the limiting-reactant lab report. Heads-up: the **Semester 1 exam** (Modules 1–4) follows this module — start reviewing now.
- 5 Gases and energy. The gas laws *describe*; kinetic-molecular theory *explains* — keep that difference in view. Don't miss the calorimetry data analysis (Excel) and the lab report. Tip: watch your units on R.
- 6 Intermolecular forces quietly explain almost every property you can see — boiling points, solubility, why ice floats. Don't miss the "why matter behaves" argument. Advance your IMF & States model.
- 7 Our biggest module: entropy, free energy, equilibrium, and acids and bases. Pace yourself with the two checkpoints. Don't miss the Le Châtelier CER. Tip: build an ICE table every time — don't shortcut it.
- 8 Electrochemistry and kinetics: redox does real work in batteries and corrosion. Don't miss the "Powering the Future" battery argument — weigh cost *and* benefit. Advance your cell build.
- 9 We close with the nucleus — decay, energy, and responsibility. Don't miss the Curie case study and the final VR video. Heads-up: the cumulative **Final exam** follows, with a nuclear cost–benefit question. You've built to this.

SITUATIONAL SCRIPTS

- **Mid-module check-in:** "If your confidence on the last reflection was low, that's useful information, not a verdict. Reply to this or grab an office-hours slot and we'll close the gap before the exam."
- **Pre-exam:** "The [Semester 1 / Final] exam opens [date]. Rework the hands-on sets and the checkpoint items — that is the exam in miniature."
- **Re-engagement:** "I haven't seen work from you in a bit and I'd like to help. Tell me where you're stuck — we'll make a plan."

10. Teacher Workflow & Workload Map

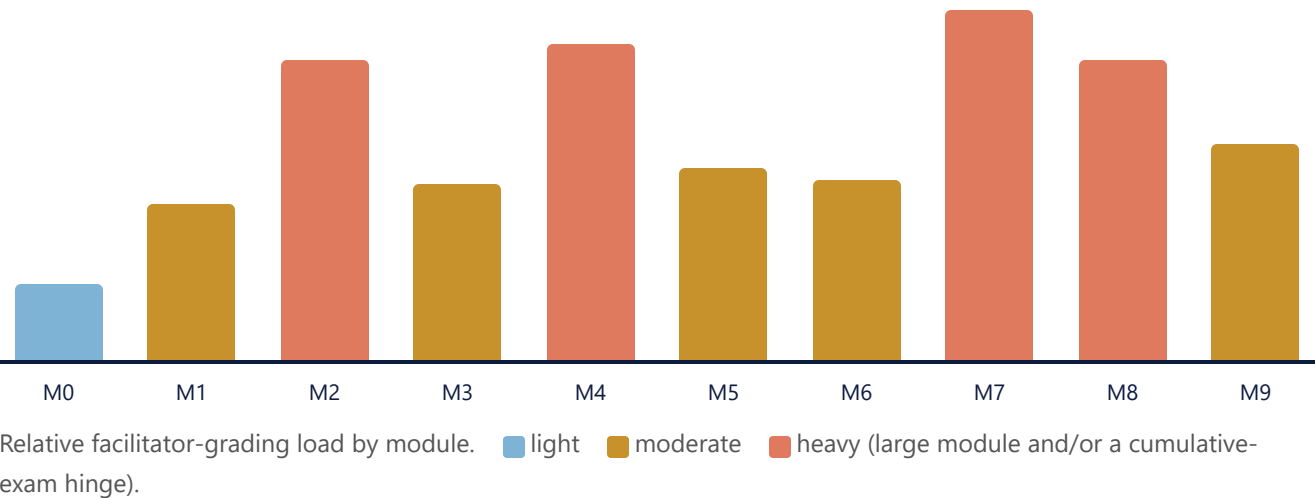
A NORMAL WEEK, IN ORDER

- 1. Triage the latest module's **outro-wrapper analytics** (15–20 min — highest leverage).
- 2. Batch-grade any **module writing** due, with rubric 6a.
- 3. Batch-grade any **VR project videos** due, with rubric 6b.
- 4. Post the **weekly announcement** (Section 9).
- 5. Answer messages and hold office hours.

Weekly time budget: roughly **0h** on Tier 1 auto work, **2–3h** on light touchpoints, and **8–12h** in the weeks a module's two Tier 3 items land — spread across the module rather than all at once.

WHERE THE HEAVY WEEKS ARE

Every module has exactly two teacher-graded items, so load is fairly even — but the large modules (more scholars, longer writing) and the two exam hinges run heavier. Plan extra time on the coral bars.



11. Standards Coverage Map (CPALMS → Modules)

[VERIFY] — proposed alignment. These codes are read from the course's four quarterly crosswalk documents and are the build's proposed alignment. Confirm against the CPALMS 2024 course description before printing in any official record. **Codes appear in this teacher copy only — never on a student page.**

11a. Quick reference — standards by module

MOD	TITLE	BENCHMARKS CARRIED (PRIMARY)
0	Getting Started	N.1.1, N.1.2, N.1.7, N.2.4, N.3.5 (Nature-of-Science only; no content benchmark)
1	Intro to Chemistry	P.8.1, P.8.2; N.1.1, N.1.5, N.1.6, N.3.1, N.3.3; MTR.3.1/.6.1/.7.1
2	Atomic Structure & Periodic Table	P.8.3, P.8.4, P.8.5, P.8.7, P.10.9, P.10.18; N.2.4, N.3.1, N.3.2, N.3.5; MTR.3.1/.5.1
3	Chemical Bonds & Geometry	P.8.5, P.8.6 (bonding half), P.8.7, P.8.12, P.8.13; N.2.5, N.3.5
4	Reactions & Stoichiometry	P.8.7, P.8.8, P.8.9 (spine), P.8.10 (intro); N.2.5, N.3.3, N.3.5; MTR.3.1/.6.1/.7.1
5	Gas Laws & Thermochemistry	P.12.10, P.10.5, P.8.9, P.10.1, P.10.2, P.10.6, P.10.7; N.3.3, N.3.5
6	Liquids, Solids, Solutions	P.8.6 (IMF half), L.18.12, P.12.11, P.8.7; N.2.5, N.3.5; MTR.3.1/.6.1/.7.1
7	Thermo, Equilibrium & Acid-Base	P.10.8, P.12.13, P.8.11, P.8.9, P.10.1/.2/.6/.7; N.3.3, N.4.1, N.4.2
8	Electrochemistry, Redox & Kinetics	P.8.10 (spine), P.10.1, P.8.9, P.12.12, P.10.5, P.10.6; N.4.1, N.4.2, L.17.15
9	Nuclear Chemistry	P.10.11, P.10.12, P.10.10, P.8.4, P.10.1/.2; N.2.4, N.4.1, N.4.2, L.17.15

11b. By standard — where each benchmark is taught & assessed

BENCHMARK	STATEMENT (ABBREVIATED)	TAUGHT (PRIMARY) · ASSESSED
P.8.1	Four states/phases of matter	M1 (1.02, 1.07) · RC, HO, CP/EX M1
P.8.2	Physical vs chemical properties & changes	M1 (1.03, 1.07) · density lab, CP/EX M1
P.8.3	Atomic model changes over time, driven by evidence	M2 (2.01, 2.02) · model-change essay + VR project, EX M2
P.8.4	Atomic structure (p / n / e)	M2 (2.03) · M9 (9.01, foundation) · RC/CP/EX
P.8.5	Atomic properties & periodic position ↔ electrons	M2 (2.08–2.10); reinforced M3 · EX M2, M3
P.8.6	Bonding forces vs other attractions (incl. H-bond, van der Waals)	Split: bonding half M3 (3.01, 3.02); IMF half M6 (6.01, 6.04, 6.05) · EX M3, M6
P.8.7	Interpret formula representations	M2 (2.04), M3 (3.03–3.06), M4 (4.01, 4.02), M6 (6.04) · RC/EX
P.8.8	Characterize reaction types (redox, acid-base, synthesis, replacement)	M4 (4.06) · VR project, CP2/EX M4
P.8.9	Mole concept + conservation → quantities (stoichiometry spine)	M4 (4.01, 4.05, 4.07, 4.08); reused M5 (5.03), M7 (7.12), M8 (8.07) · HO/EX
P.8.10	Redox in living & non-living systems	Introduced M4 (4.06); developed fully M8 (8.01–8.07) · EX M4, M8
P.8.11	Acidity/basicity ↔ $\text{H}_3\text{O}^+/\text{OH}^-$ & pH	M7 (7.07–7.09) · RC/HO/CP/EX M7, VR project
P.8.12 / P.8.13	Carbon's versatility; functional groups (organic fold-in)	M3 (3.07); properties reinforced M6 (6.01) · RC, EX M3
P.10.1 / P.10.2	Forms of energy; conservation of energy	M5 (5.07); M7 (7.03); M8 (8.02); M9 (9.04, $E=mc^2$) · HO/EX
P.10.5	Temperature ↔ average molecular KE	M5 (5.05); M8 (8.09) · RC/EX
P.10.6 / P.10.7	Potential-energy diagrams; endo/exothermic	M5 (5.07, 5.08); M7 (7.01); M8 (8.09, 8.10) · RC/EX

BENCHMARK	STATEMENT (ABBREVIATED)	TAUGHT (PRIMARY) · ASSESSED
P.10.8	Entropy & efficiency of energy-to-work	M7 (7.01–7.03) · RC/HO/CP/EX M7, VR project
P.10.9	Quantization of energy at the atomic level	M2 (2.06) · RC/CP/EX M2
P.10.10	Four fundamental forces (strong/weak nuclear)	M9 (9.01) · RC/CP/EX M9
P.10.11	Nuclear reactions (decay/fission/fusion), energy, safety	M9 (9.02–9.06) · RC/HO/CP/EX M9, capstone + VR project
P.10.12	Differentiate chemical vs nuclear reactions	M9 (9.01) · RC/CP/EX M9
P.10.18	EM spectrum: wavelength / frequency / energy	M2 (2.05) · RC/HO/CP/EX M2
P.12.10	Ideal gases via kinetic-molecular theory	M5 (5.01, 5.02, 5.05) · RC/HO/CP/EX M5, VR project
P.12.11	Phase transitions via KMT	M6 (6.03) · RC/HO/CP/EX M6
P.12.12	Factors affecting reaction rate	M8 (8.08–8.10) · RC/HO/EX M8, VR project
P.12.13	Dynamic equilibrium (equal-rate reversible processes)	M7 (7.04–7.06) · RC/HO/CP/EX M7, CER + VR project
L.17.15 / L.17.19	Technology & environmental quality; natural-resource use	M8 (batteries), M9 (nuclear power) · battery argument, capstone
L.18.12	Special properties of water	M6 (6.02, 6.06) · RC/CP/EX M6
N.1.x	Problem-defining, observation, source reliability, reproducibility, inference	Woven all year; anchored M0, M1, M2; N.1.4 via the AI-as-Instrument thread
N.2.4 / N.2.5	Knowledge durable yet revisable; competing interpretations a strength	M2 (2.01), M3 (3.02), M4 (4.05), M6 (6.05), M7 (7.07), M9 (9.01)
N.3.1–N.3.5	Theory vs law; consensus; function & limits of models	M1 (1.01), M2, M4 (4.05), M5 (law vs KMT), M7–M8 · NoS thread

BENCHMARK	STATEMENT (ABBREVIATED)	TAUGHT (PRIMARY) · ASSESSED
N.4.1 / N.4.2	Empirically-informed societal decisions; weighing costs & benefits	M7 (Le Châtelier CER), M8 (battery argument), M9 (nuclear cost/benefit)
MTR.3.1/.5.1/.6.1/.7.1	Math practices: fluency, patterns, reasonableness, real-world application	Every quantitative week; heaviest M1, M4, M5, M7, M9
EE.1.1 / EE.4.1	Cite evidence; academic argument with counterclaims	Every CER, lab report, and scientific argument (esp. M2, M3, M7, M8)

Deliberate design notes (not gaps): molarity/concentration (M4) and colligative properties (M6) have no dedicated CPALMS code — they ride on P.8.9 + the math MTRs. Gibbs free energy (M7), enthalpy/bond energy (M5), electrochemical cells (M8), and half-life (M9) likewise have no standalone code and are carried by the P.10.x energy cluster, P.8.10, and the MTRs respectively. P.8.6 is fully retired across M3 + M6.

12. What Makes This Course Classical & Cultivating Culture

What makes it classical

- **Chemistry as a centuries-long investigation.** Scholars meet the discipline through the questions and experiments that built it — Democritus, Lavoisier, Dalton, Mendeleev, the Curies, Rutherford, Bohr — not as a finished list of facts.
- **The four running threads.** Empirical Questioning (every lesson moves phenomenon → evidence → explanation), Nature of Science (every quarter revisits *how do we know?*), Modern Research (*Then* → *Now* → *Future* in every module), and AI-as-Instrument (with human verification) are the classical spine.
- **Wonder before mechanism.** Lessons open on a real phenomenon and let the model earn its place, rather than front-loading the conclusion.
- **A virtue undercurrent.** Q1 Wonder, Q2 Perseverance, Q3 Self-Government, Q4 Responsibility — with Honesty (evidence over preference) running through all four via the Nature-of-Science thread.

Cultivating culture asynchronously

- Use the weekly announcement to model curiosity, not just logistics — name a phenomenon worth wondering about.
- In feedback and any public comments, name scholars' good *reasoning*, not just right answers.
- Keep the virtue thread visible: a line about honest data or perseverance through a hard derivation costs nothing and sets the tone.
- Treat office hours and messages as the relationship that carries an async class.

13. Sensitive-Content Flags & Family/Student Scripts

Chemistry is low-sensitivity, but three areas warrant an advance word. Language below is neutral and bracketed for your voice and contact info.

WHERE	WHAT IT IS	SUGGESTED ADVANCE SCRIPT
Module 0 & throughout — lab safety	Safety practices and hazard awareness; all labs are virtual or data-based (no home reagents)	"This course does its lab work in VR and with data — scholars are not asked to handle chemicals at home. We still teach real lab-safety habits, because thinking safely is part of thinking like a scientist."
Module 9 — nuclear chemistry	Radioactivity, radiation's biological effects, and the history of nuclear energy and weapons; a cost–benefit capstone	"In our final module we study the nucleus — including radiation safety and the history and trade-offs of nuclear technology. We treat it as scientists and citizens do: weighing real benefits (medicine, energy) against real risks. Reach out with any questions."
Modules 7–9 — societal debates	Ocean acidification, green chemistry, battery materials, nuclear power — topics with policy dimensions	"Some assignments ask scholars to weigh evidence on real issues like ocean acidification or energy choices. We keep the focus on the chemistry and on reasoning from evidence, not on partisan positions."

14. Adapting the Async Course to a Live Class That Meets Twice a Week

The same Canvas content runs unchanged in a live section — you simply pull the week's investigative arc into two class meetings and leave the auto-graded pieces as homework. Here is a repeatable weekly pattern, then what to prioritize live in each module.

The repeatable two-day week

WHEN	IN THE LIVE MEETING (~50–60 MIN)	ASSIGNED AROUND IT
Day 1 — Anchor	Open on the week's <i>phenomenon</i> and the Grand Question. Deliver the core lesson(s) as a short narration, running the embedded interactive together (Build-an-Atom, the equation balancer, the ICE-table solver). Work one example of the week's hands-on problem type at the board.	Before Day 2: read the remaining lesson page(s), take those Reading Checks, and start the weekly hands-on set.
Day 2 — Do chemistry	Run the week's VR activity live as a class (shared build or gallery walk), or do the lab/data task together. Then review the hands-on set — surface the common wrong answers — and give scholars work time to advance the module's cumulative VR project increment. Close by previewing next week.	After Day 2: finish the hands-on set and the VR increment; keep the module writing moving on its due cadence.

HOLDS TRUE EVERY WEEK

- **Reading Checks are always homework** — they are the pre-class accountability that frees class time for reasoning and lab work.
- **The two teacher-graded items keep the module cadence** — the writing assignment and VR project video are due at module's end just as in the async course; use Day 2 work-time to keep them moving.
- **Two-lesson (calc-heavy) weeks → more board time.** Spend Day 1 almost entirely on worked problems and give the interactive as guided practice.
- **Three-lesson (descriptive) weeks → more discussion.** You can cover two lessons in narration on Day 1 and leave the third as reading, using Day 2 for the VR/gallery activity and a Nature-of-Science discussion.

What to prioritize live, module by module

MOD	BEST USE OF LIVE TIME
0	Norm-set the semester: run OptimaXR together, practice one CER aloud, model the science notebook. One meeting is usually enough.
1	Calc-heavy back half: put significant figures and dimensional analysis on the board; do the density measurement live as the lab.
2	Longest module — run it as the atom's argument from evidence. Day 1 narrate the model's evolution; Day 2 do Build-an-Atom and the atomic-history gallery walk.
3	Draw Lewis structures and predict VSEPR shapes together at the board — this is hard to do alone. Use VR for the geometry lab.
4	All calculation: balance, then mole-bridge, then limiting reactant — live worked examples every meeting. Reserve a Day 2 for Semester 1 exam review.
5	Gas-law and calorimetry problems on the board; run the gas-law manipulator and calorimetry setup live (or the Excel data task together).
6	Mostly descriptive: Day 1 IMF reasoning and boiling-point ranking; Day 2 crystal-lattice / dissolution build and colligative calculations.
7	Biggest module — budget both days most weeks. Build ICE tables together; walk a titration live; discuss the Le Châtelier / ocean-acidification case before the CER.
8	Build a galvanic cell together and trace electron flow; put cell-potential and rate-law calculations on the board.
9	Balance nuclear equations and work half-life live; devote a discussion to the nuclear cost–benefit question before the capstone, then Final exam review.

One rule of thumb: use live time for what scholars cannot do alone — worked calculations, hands-on labs, and evidence-weighting discussion. Leave reading and the auto-graded checks for independent work. The async course already carries the content; your two meetings add the reasoning and the community.